Jeongho Son

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Education

Pohang University of Science and Technology (POSTECH) Pohang, South Korea
B.S. in Computer Science; GPA: 3.88/4.30 *Feb. 2021 – Expected Feb. 2028*

Sejong Science High School *Feb. 2018 – Feb. 2021*

Projects

Optimization Grand Challenge 2026 | *C++20, Python, OpenMP, pybind11* June 2026 – Sept. 2026

- Led team 최적화야호 to the **Expert Optimizer award (3rd Prize; KRW 10,000,000)** among 958 teams.
- Designed a geometry engine with slabs and bit-packed tables: 2,407–12,769× faster than naive vectorized Shapely.
- Combined adaptive large neighborhood search (ALNS) with cached graph reachability and the geometry engine to optimize shipyard block placement and scheduling while enforcing crane-operation precedence.

Distributed Sorting | *Scala, gRPC* Sept. 2023 – Nov. 2023

- Built a distributed external sorting system for datasets exceeding a single server's disk capacity; validated sorting of 40 GB (400 million records) across four workers.
- Ranked **first in performance** and **first in design simplicity** among three CS434 project teams.

Standard ML Subset Compiler | *OCaml* Apr. 2023 – June 2023

- Implemented a compiler backend for Tiny ML, a Standard ML subset, generating assembly for the Mach abstract machine with support for higher-order functions, recursion, and pattern matching.
- Built heap-allocated closures to capture lexical environments; used constant folding and compile-time variable bindings to reduce runtime instructions and stack operations.

POSTECH–KAIST AI Competition (포카전) | *Python, PyTorch* Mar. 2024 – Sept. 2024

- Our POSTECH team **won the AI competition against KAIST** with a Super Smash Bros. agent.
- Our team developed the agent in Python and PyTorch using proximal policy optimization (PPO), grouping consecutive moves into composite actions.

PSPNet Reproduction and Evaluation | *Python, PyTorch* Oct. 2023 – Dec. 2023

- Modified PSPNet's pyramid pooling to share context across levels; trained both variants on PASCAL VOC 2012.

Research Experience

POSTECH Computer Vision Laboratory Sept. 2024 – Dec. 2024

- Studied activity grammars for temporal action segmentation.

POSTECH Data Systems Laboratory Jan. 2024 – Feb. 2024

- Studied Transformer architectures, including attention, positional encoding, and encoder–decoder design; documented concepts in research notes.

Honors and Service

Student Mentoring Program (SMP) Feb. 2022 – Dec. 2024

- Tutored students in physics, computer science, and algorithms.

National Science & Technology Scholarship (이공계 국가우수장학금) Aug. 2023

- Awarded **full tuition through graduation** for academic achievement.

POSTECH Global Leadership Program (GLP) Mar. 2023

- Received a **KRW 2,500,000 university scholarship** for academic achievement.

Technical Skills and Languages

Programming Languages: C++, Python, Scala, OCaml
Libraries and Tools: PyTorch, OpenMP, pybind11, gRPC, Git
English: TOEFL iBT 5.5/6.0 (110/120), Aug. 8, 2026; TEPS 527/600 (Level 1+), July 5, 2025